



Background 01

20–30% of acute pancreatitis (AP) admissions progress to moderate or severe disease with organ failure or necrosis. Existing bedside scores (Ranson, APACHE II, BISAP) are validated but coarse, and most published ML severity models are never released as usable tools — no implementation another center can test against its own patients.

PenuX packages a trained severity model as a self-contained, inspectable, freely runnable tool — built specifically to be externally validated, not just cited.

In-browser inference pipeline 02

The model — a 200-tree random forest over 106 admission-window features — runs as genuine CPython via **Pyodide** (real scikit-learn-style tree traversal compiled to WebAssembly), not a hand-ported JS approximation. No patient data is transmitted for scoring.



Cohort & status 03

722	106	200	.535
PATIENTS	FEATURES	TREES	THRESHOLD

Calibration curves, Brier score, and lognormal-fit diagnostics are computed live in-browser from raw model outputs — openly inspectable, not a static report.

No prospective, multi-site external validation exists yet. That gap is the reason for this poster.

New: in-browser model compression 04

A companion tool prunes tree depth and tree count of the same 200-tree model — entirely client-side — then checks how closely the compressed model still agrees with the full one, using synthetic inputs sampled across each feature's real threshold range.

73%	0.84	81%
smaller file — 50 trees, depth ≤ 6	correlation to full-model output	severe / not-severe agreement

Fidelity-to-original-model figures on synthetic inputs — a self-consistency check, not a clinical accuracy claim.

Collaboration sought 05

- External validation on a held-out or prospective cohort — patient data never has to leave your site.
- Clinical review of feature choices, decision threshold, and triage fit.
- Open to co-authorship on a validation write-up, or a citation if useful as-is.